

Why Matrix SYK Models Do Not Have R-Charge Concentration

Abstract. R-charge concentration — all BPS states of an irreducible complex sitting in a single R-charge — is the proposed fingerprint of a fortuitous, black-hole-like BPS sector, and Chen has conjectured that purely fermionic $U(N)$ matrix quantum mechanics with a cubic supercharge and three flavours realises it. We compute the BPS cohomology of these models exactly. The $U(N)$ weight decomposition splits the Q -complex into pieces of dimension $\leq 10^4$ in place of 2^{pN^2} , which puts $N \leq 5$ within reach of exact integer linear algebra; Kostant inversion converts weight-space cohomology into per-irrep BPS counts. The results are uniformly negative and follow a simple law. The BPS states occupy a window of *consecutive* R-charges whose width grows linearly with the gauge rank: $W = N + 1$ for $p = 1, 3$, $W = 2N + 1$ for $p = 2, 4$, $W = 3N + 1$ for $p = 5$, with period four in p . Since the complexes are graded by $N_\Psi \bmod q$ with $q = 3$ fixed, a window of width $W > q$ cannot fit one degree per class, and concentration fails for every $N \geq 3$. We then solve the problem analytically: at the maximal weight the complex factorises over the rank G Cartan directions into $(\Lambda^\bullet \mathbb{C}^p, \omega \wedge)^{\otimes \text{rank } G}$, giving $W = \text{rank } G (w_s - 1) + 1$ in closed form, and an Euler-characteristic argument forces $w_s \geq 2$. Concentration therefore requires $\text{rank } G \leq q - 1$, and solving for w_s sharpens this to $\text{rank } G \leq 2$ — exactly $u(2)$ and $su(3)$. Chen’s supporting $N = 2$ evidence is one of the only two cases in the entire family where concentration holds, and it holds for kinematic reasons.

1 Introduction

The models. The degrees of freedom are complex fermionic $N \times N$ matrices Ψ_{ij}^a , $a = 1, \dots, p$, with $\{\Psi_{ij}^a, \bar{\Psi}_{kl}^b\} = \delta^{ab} \delta_{il} \delta_{jk}$ and $U(N)$ acting by conjugation. All dynamics comes from a nilpotent supercharge,

$$Q = \sum_{a_1 \dots a_q} C_{a_1 \dots a_q} \text{Tr}[\Psi^{a_1} \dots \Psi^{a_q}], \quad H = \{Q, \bar{Q}\} \geq 0, \quad [N_\Psi, Q] = qQ, \quad (1)$$

with $N_\Psi = \text{Tr} \Psi^a \bar{\Psi}^a$ the R-charge. Only odd q is admissible: $\text{Tr}[\Psi^q]$ picks up $(-1)^{q-1}$ under a cyclic shift, so it vanishes identically for even q , and $Q^2 = 0$ holds automatically precisely when q is odd, because exchanging the two q -tuples in $C \otimes C$ is a permutation of $2q$ fermions with q^2 transpositions, contracted against a totally antisymmetric product. The case $q = 3, p = 1$ is Chen’s exactly solvable single-matrix model [1], where H is a function of the gauge Casimir; $q = 3, p = 3$ with Chen’s couplings is the proposed “matrix SYK” model, for which the Casimir structure is lost.

BPS states are $E = 0$ states, equivalently classes in $\ker Q / \text{im } Q$. Because Q is a gauge singlet and raises N_Ψ by q , the cohomology splits into irreducible complexes labelled by the $U(N)$ irrep λ , the $\mathbb{Z}_p$ flavour charge ω , and the residue $c = N_\Psi \bmod q$.

Concentration, and what is at stake. Chang, Chen, Sia and Yang [2] propose *R-charge concentration* as the signature of a fortuitous BPS sector: along each irreducible complex with macroscopic index, all BPS states occupy a *single* degree. In $\mathcal{N} = 2$ SYK this holds, the BPS counts follow the $\cos(\pi k/3)$ profile of the $\mathcal{N} = 2$ super-Schwarzian [5, 4], and the index counts the BPS states exactly. Chen observes that his single-matrix model fails the test — its BPS states span $N + 1$ consecutive charges [1, (2.22)] — and conjectures that with three flavours “the magic number might be three”, citing an unpublished $N = 2$ simulation that “suggests” concentration. Whether this survives $N \geq 3$ has been open, and it is the question we settle.

Summary of results.

- The BPS window is a run of consecutive R-charges centred on half filling, of width W growing linearly in N : $W = N + 1$ ($p = 1, 3$), $2N + 1$ ($p = 2, 4$), $3N + 1$ ($p = 5$), with period four in p (Table 1). Exact counts at $N = 2, 3, 4, 5$ (Table 2).
- Since the grading period is $q = 3$, $W > 3$ makes one-degree-per-class impossible. Concentration holds at $N = 2$ and fails for every $N \geq 3$ — by counting, not by dynamics.
- Going traceless buys exactly one unit: $su(N)$ has window N , so $su(3)$ concentrates and $su(4)$ does not. The controlling parameter is $\text{rank } G$, not N .
- Analytically, the maximal-weight complex factorises over Cartan directions, $Z_{\max}(t) = t^{k_0} (z_{\text{slot}}(t))^{\text{rank } G}$, giving $W = \text{rank } G (w_s - 1) + 1$; an Euler argument gives $w_s \geq 2$, hence $\text{rank } G \leq q - 1$, and solving for w_s gives $\text{rank } G \leq 2$.

p	1	2	3	4	5	6	7	8
w_s	2	3	2	3	4	3	2	3
window width W	$N+1$	$2N+1$	$N+1$	$2N+1$	$3N+1$	$2N+1$	$N+1$	$2N+1$

Table 1: BPS window width for $q = 3$ as a function of the number of flavours, from the closed form $W = \text{rank } G(w_s - 1) + 1$ of Sec. 4 with $\text{rank } U(N) = N$. The pattern has period four in p : $W = N + 1$ for $p \equiv 3 \pmod{4}$ and $p = 1$, $2N + 1$ for even p , $3N + 1$ for $p \equiv 1 \pmod{4}$, $p > 1$. Verified directly at $N = 3$ for $p = 1, 2, 3, 4$. In every case $W > q = 3$ once $N \geq 3$.

2 Methodology

Weight-space decomposition. Direct computation of $\ker Q / \text{im } Q$ is hopeless: at $N = 3$, $p = 3$ the middle sector has $\dim \mathcal{H}_{13} = 2 \times 10^7$, and at $N = 4$ it is 3×10^{13} . The observation that removes the problem is that Q in (1) is a $U(N)$ singlet — every term is a closed index loop — so it commutes with the Cartan and preserves the $U(N)$ weight. The complex therefore splits over weights,

$$\cdots \xrightarrow{Q} W_\lambda(k) \xrightarrow{Q} W_\lambda(k+q) \xrightarrow{Q} \cdots, \quad \dim H^k(W_\lambda) = \dim W_\lambda(k) - \text{rank } Q_k - \text{rank } Q_{k-q}, \quad (2)$$

and the weight spaces are small: the largest relevant one is 126 at $N = 3$ and 3×10^4 at $N = 4$. Since Ψ_{ij}^a carries weight $e_i - e_j$, a state's weight depends only on the occupation matrix n_{ij} , and enumerating $W_\lambda(k)$ is a pruned search over n_{ij} rather than a scan of Fock space.

Exact ranks. Ranks are computed by Gaussian elimination over $\mathbb{F}_P$. A rank over $\mathbb{F}_P$ can only under-estimate the rational rank, so every value reported here was recomputed over a second prime and, where feasible, in floating point. For the larger weight spaces we use a right-looking blocked LU whose trailing update is a single BLAS matmul; exactness is preserved by choosing $P \approx 2^{20}$ so that $\text{block} \times (P - 1)^2 < 2^{53}$ and every float64 partial sum is an exact integer. A further factor of p^2 in time and memory comes from the $\mathbb{Z}_p$ flavour symmetry: Q commutes with the flavour rotation, so each weight space splits into p blocks of size $\approx n/p$, written exactly over $\mathbb{F}_P$ with $P \equiv 1 \pmod{p}$ so that a primitive p -th root of unity exists. Together these take the largest tractable weight space from 924 to 30 294; the complete $N = 4$ run below takes 30 minutes at 6 GB.

From weights to irreps. A weight space collects contributions from every irrep containing that weight,

$$\dim H^k(W_\lambda) = \sum_{\mu \geq \lambda} K(\mu, \lambda) h_\mu^k, \quad K(\mu, \lambda) = \sum_{w \in W} \text{sgn}(w) \mathcal{P}(w(\mu + \rho) - (\lambda + \rho)), \quad (3)$$

with K the Kostant weight multiplicity and $\mathcal{P}$ the Kostant partition function. The system is triangular and is inverted from the top down. Crucially it is not a truncation: higher weights have *smaller* weight spaces, so the affordable set is automatically closed upward.

Validation. Every number below passes four independent checks: (i) the weight-space dimensions reproduce the multiplicities obtained from the Fock-space character by a Weyl alternating sum; (ii) for each irrep and each class c , the Euler characteristic $\sum_{k \equiv c} (-1)^k h^k$ equals the independently computed refined Witten index, including when refined by flavour charge; (iii) all peeled multiplicities are non-negative, a strong constraint since the peel subtracts large numbers; (iv) the particle-hole symmetry $h^k = h^{pN^2 - k}$ emerges without being imposed. At $N = 2$ the method reproduces exact diagonalisation, and for $p = 1$ it reproduces Chen's closed form [1, (2.21)–(2.22)] at $N = 2, 3, 4$.

3 Results

The window and its width. For every model and every irrep we have computed, the BPS cohomology occupies a run of *consecutive* R-charges centred on half filling $pN^2/2$. The width obeys a simple law, periodic in p with period four (Table 1), and grows linearly in N in all cases. Table 2 gives the exact multiplet counts for the maximal irrep at $p = 3$; the profiles are binomial, $3^N \binom{N}{j}$, and the totals are 6^N .

Why concentration fails, and why $N = 2$ is misleading. The complexes are graded by $c = N_\Psi \pmod{3}$. A window of W consecutive charges distributes over the three classes with $\lceil W/3 \rceil$ charges in the most populated one, so one-degree-per-class requires $W \leq 3$. At $N = 2$, $W = 3$ exactly: each class receives exactly one charge and concentration holds — for purely kinematic reasons, with no input from the dynamics. This is the regime of the $N = 2$ evidence cited for the matrix-SYK proposal. From $N = 3$ it is arithmetically impossible. At $N = 3$ the surplus falls in the $c = 0$ class, whose index vanishes, so the extra states are invisible to the index and the non-zero-index classes still saturate; at $N = 4$ the surplus falls in $c = 1, 2$, whose index does *not* vanish, so saturation fails where

N	window (N_Ψ)	width	BPS multiplets h^k	profile
2	5, 6, 7	3	9, 18, 9	$3^2 \binom{2}{j}$
3	12, ..., 15	4	27, 81, 81, 27	$3^3 \binom{3}{j}$
4	22, ..., 26	5	81, 324, 486, 324, 81	$3^4 \binom{4}{j}$
5	35, ..., 40	6	243, 1215, 2430, 2430, 1215, 243	$3^5 \binom{5}{j}$

Table 2: Exact BPS multiplet counts for the maximal irrep of the three-flavour model, $\lambda_i^* = p(N+1-2i)$. The window is $N+1$ consecutive charges centred on $pN^2/2$, and $Z_{\text{BPS}} = 3^N(1+t)^N t^{k_0}$, i.e. 6^N multiplets. The $(1+t)^N$ factor is identical to Chen’s single-matrix result [1, (2.20)] — the model he identifies as *not* concentrated.

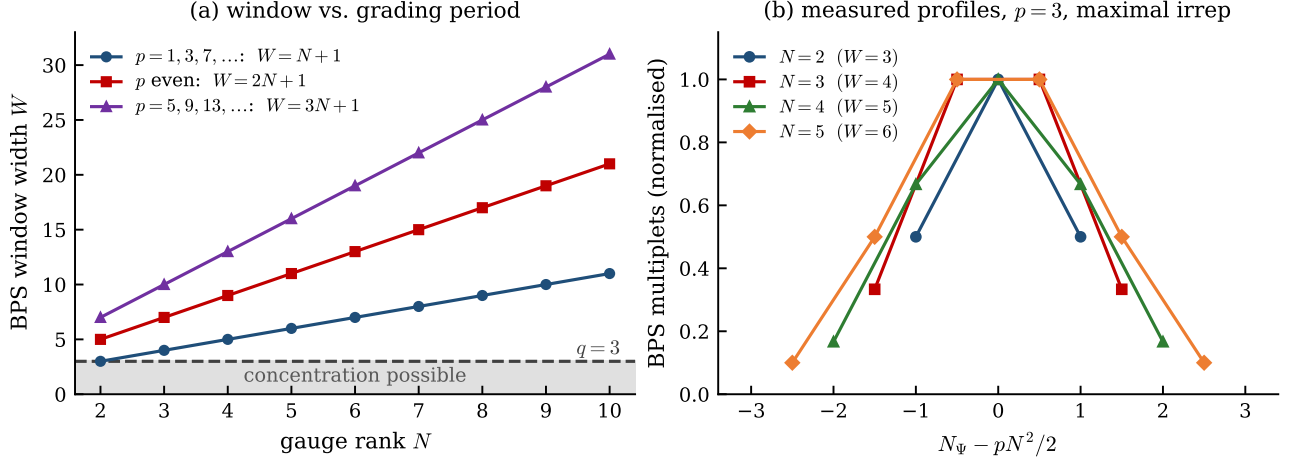


Figure 1: (a) BPS window width against gauge rank. The grading period $q = 3$ is fixed (dashed), so concentration is possible only inside the shaded band; every flavour number leaves it at or before $N = 3$. (b) Measured BPS multiplet profiles for $p = 3$ at the maximal irrep, normalised and centred on half filling: the window visibly widens by one charge per unit of N .

the index is macroscopic. Across the thirteen highest irreps at $N = 4$ we count 3,008,825,568 BPS states of which 19.0% are invisible to the index.

Robustness. The conclusion does not depend on the couplings, the gauge group, or the supercharge degree. (i) *Couplings:* sampling 30 random cyclic tensors C at $N = 3$, $p = 3$, 26 give Chen’s profile exactly and 4 give a *wider* window (width 10), never a narrower one. (ii) *Casimir:* descending the spectrum at $N = 3$ through complete Casimir tiers from $C_2 = 48$ to $C_2 = 24$ — thirteen irreps, up to Fock multiplicity 110 080, a factor 215 above the maximal irrep — the support is width 4 in every single case. (iii) *Gauge group:* $su(N)$ removes one Cartan direction and narrows the window by exactly one, so $su(3)$ (rank 2) concentrates with $h = 9, 18, 9$ while $su(4)$ (rank 3) does not; the controlling parameter is rank G . (iv) *Degree:* raising q does not help, for the reason given next.

4 The analytic no-go

Factorisation over Cartan slots. At the maximal weight $\lambda_i^* = p(N+1-2i)$ every off-diagonal mode is saturated, so the only unoccupied modes are the diagonal ones. A gauge-invariant Q must create a closed index loop, and any loop entering an index cannot leave it — its outgoing edges are occupied. The surviving loops are therefore *self-loops* at a single index, creating q diagonal modes of distinct flavours. Hence $Q = \sum_{i=1}^{\text{rank } G} Q_i$ with Q_i acting only within slot i , and Q_i is wedging with the totally antisymmetric part of C . By Künneth,

$$Z_{\text{max}}(t) = t^{k_0} (z_{\text{slot}}(t))^{\text{rank } G}, \quad z_{\text{slot}}(t) = \sum_j \dim H^j(\Lambda^\bullet \mathbb{C}^p, \omega \wedge) t^j, \quad \omega = \text{Alt}(C) \in \Lambda^q \mathbb{C}^p. \quad (4)$$

Equation (4) reproduces all ten measured $u(N)$ cases exactly, across $p = 1, 2, 3, 4$, $q = 3, 5$ and $N = 2, \dots, 5$. If z_{slot} has support on w_s degrees then the window width is $W = \text{rank } G (w_s - 1) + 1$, which is the law in Table 1.

$w_s \geq 2$, and the bound. Let $E_c = \sum_{j \equiv c(q)} (-1)^j \binom{p}{j}$ be the Euler characteristic of the class- c subcomplex of $\Lambda^\bullet \mathbb{C}^p$; $E_c \neq 0$ forces cohomology somewhere in class c . Two elementary facts pin this down: $\sum_c E_c = (1-1)^p = 0$ for $p \geq 1$, so exactly one non-zero E_c is impossible; and $E_c = \frac{1}{q} \sum_{m=1}^{q-1} \zeta^{-mc} (1-\zeta^m)^p$ with $\zeta = e^{2\pi i/q}$ and $1-\zeta^m \neq 0$, so they cannot all vanish. At least two classes therefore carry cohomology, $w_s \geq 2$, and

$$W \geq \text{rank } G + 1 \quad \implies \quad \text{concentration} \implies \text{rank } G \leq q - 1. \quad (5)$$

Raising q cannot rescue this. With $\omega \neq 0$ (which requires $q \leq p$) the minimum of w_s is $q - 1$ for $q \geq 5$, so $\text{rank } G \leq \lfloor (q - 1)/(q - 2) \rfloor = 1$; only $q = 3$ attains $w_s = 2$, and there $\text{rank } G \leq 2$. Scanning all $q \leq p \leq 14$, the maximum rank compatible with concentration is **2**, attained at $q = 3$ with $p \equiv 3 \pmod{4}$. Scaling p yields infinitely many concentrating models, all at rank 2.

The free branch. One evasion exists and is instructive. If $q > p$ then $\omega \in \Lambda^q \mathbb{C}^p = 0$, the differential vanishes on the sector, $z_{\text{slot}} = (1 + t)^p$, and for $p = 1$ this gives $w_s = 2$ with arbitrary q , hence unbounded rank. But $\omega = 0$ means Q annihilates the sector outright: every state is BPS because nothing acts on it, not because of fortuity. This is the single-matrix, Casimir-solvable situation. It satisfies the counting definition of concentration while being physically its opposite, which shows that the counting criterion alone is not sufficient — one needs Q to act non-trivially *and* the window to fit inside one period, and (5) says those demands are incompatible beyond rank 2.

5 Outlook

- **Close the last gap in the theorem.** The bound $\text{rank } G \leq q - 1$ is proved; the sharpening to $\text{rank } G \leq 2$ uses the observed $w_s^{\min} = q - 1$ for $q \geq 5$, verified at $q = 3, 5, 7, 9$ and by an exhaustive scan of $q \leq p \leq 14$. Proving $w_s^{\min} = q - 1$ is a self-contained question about $H(\Lambda^\bullet \mathbb{C}^p, \omega \wedge)$ with no physics in it.
- **Beyond the maximal weight.** The factorisation (4) is derived at the maximal weight and for $u(N)$; it fails for $su(N)$, where Cartan directions straddle slots, which is visible in the anomalous $su(4)$ profile 13, 81, 81, 13. Empirically the window matches the maximal one throughout the $N = 3$ spectrum down to $C_2 = 24$, but this is not derived.
- **The low-Casimir bulk.** The macroscopic-index complexes lie at low Casimir — at $N = 4$ the largest multiplicity is 6.6×10^9 at $C_2 = 20$ — where weight spaces reach 10^9 and a genuinely sparse rank routine would be needed. Our coverage spans a factor 215 in multiplicity at $N = 3$ but not the final factor of ~ 5 to the peak.
- **Chaos without concentration.** The three-flavour H is *not* a Casimir function, unlike the single-matrix case, so the model may still be chaotic and have a super-Schwarzian near-BPS spectrum even though its BPS sector is not concentrated. Nothing here bears on that, and it is the natural next question — Altland–Zirnbauer statistics of the singular values of Q_k , now cheap to extract weight by weight.
- **What would concentrate.** Equation (5) says one needs q growing with the rank, and $\omega \neq 0$ needs $p \geq q$: an interaction touching $O(N)$ of the N^2 matrix entries. Whether such a model is chaotic, or has a sensible large- N limit, is open.

6 Numerical implementation

All computations are exact integer linear algebra; no floating-point tolerance enters any reported number. Weight bases are enumerated by a pruned recursion over the occupation matrix, validated by $\sum_\lambda \dim W_\lambda(k) = \binom{pN^2}{k}$ for every k at $N = 3$ and against the Fock character at $N = 4$. Ranks use Gaussian elimination over $\mathbb{F}_P$ with $P = 2^{31} - 1$ for small blocks, and for large ones a blocked LU with the trailing update as a BLAS matmul over $P \approx 2^{20}$, chosen so that $\text{block} \times (P - 1)^2 < 2^{53}$ keeps every float64 partial sum an exact integer; every rank is confirmed over a second prime. The $\mathbb{Z}_p$ flavour blocking is performed over $\mathbb{F}_P$ with $P \equiv 1 \pmod{p}$, the eigenbasis built from orbits of the rotation with their fermionic signs, and is validated by requiring the block ranks to sum to the dense rank for every k . Kostant multiplicities are computed from the Weyl alternating sum with a pruned partition function and verified against $\sum_\lambda K(\mu, \lambda) = \dim \mu$. Representative costs: the eight highest irreps at $N = 3$ in under two minutes at 1.0 GB; thirteen irreps at $N = 4$ in 30 minutes at 6.1 GB; the $N = 5$ maximal weight in 49 seconds at 0.4 GB.

References

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